Hermes Sidecar Strategy for Trade AI

Version 4 - Research, Memory, Challenge, and Near-24/7 Self-Learning Design

Prepared for Trade AI v12 / OpenClaw project

Core decision: Trade AI remains the source of truth and execution authority. Hermes becomes the research desk, second brain, memory layer, and independent challenger. Hermes does not place trades or mutate core trading state.

Document purpose: Strategic design guide for integrating Hermes as a near-24/7 research, memory, and challenge layer for Trade AI.

Design status: Planning and architecture document. This is not an implementation approval and does not authorize trading, database mutation, broker activity, or model-routing changes.

Core decision: Trade AI remains the system of record and execution authority. Hermes becomes the research desk, second brain, durable memory layer, challenge reviewer, and strategy-learning system.

1. Executive Summary

Hermes should not be installed as the standalone Railway trading worker described in the original onboarding prompt. That pattern is useful for a new isolated trading bot, but Trade AI already has the proposal lifecycle, backtesting, journal, Command Center, ATM safety gates, local model routing, self-healing, and Drive-synced documentation.

Hermes should instead be integrated as a sidecar intelligence system:

```
Trade AI = source of truth, proposal engine, execution governance, journal, backtesting, safety.

Hermes = research desk, second brain, challenge layer, long-term memory, experiment recommender.

Claude Code = implementation mechanic under explicit operator-approved prompts.

John = final approval authority for strategy, execution, DB changes, cron changes, and broker behavior.
```

Hermes should operate close to 24/7, but not by running heavy models continuously. It should use lightweight polling and routing during market hours, normal local analysis for daily work, and off-hours deep review for large research or monthly synthesis.

The v4 design expands Hermes from a trade-review agent into a full research organization that covers:

- Tickers
- News
- Related news
- YouTube transcripts
- Earnings transcripts
- Incubator and percolator ideas
- Proposals and missed opportunities
- All trades, paper and real
- Portfolio holdings and rotation
- Retirement accounts
- Tax lots and tax-loss watchlists
- Rebalance planning
- Internal tickets and system issues
- Dashboard truth and UI comprehension

- Data freshness and source credibility
- Strategy experiments and long-term lessons

Hermes must learn from history by recording every recommendation, every operator decision, and every later outcome.

2. Strategic Positioning

2.1 What Hermes Is

Hermes is the always-learning second brain for Trade AI. It reads, researches, remembers, challenges, and recommends.

Hermes should behave like a small analyst team that never forgets:

- A ticker analyst
- A news analyst
- A transcript analyst
- A proposal challenger
- A trade postmortem reviewer
- A portfolio rotation researcher
- A retirement and tax research assistant
- A systems analyst
- A memory librarian
- A chief coordinator

2.2 What Hermes Is Not

Hermes is not:

- A broker
- A trading bot
- A replacement for Trade AI
- An approval engine
- A proposal mutation engine
- A cron editor
- A database writer for core trading tables
- A second execution path
- A system allowed to bypass safety gates

2.3 Core Rule

Hermes may recommend. Trade AI and John decide.

3. Source Grounding and Current Trade AI Baseline

The current Trade AI platform already includes:

- 23 strategy definitions and a strategy playbook
- Agent capabilities and OpenClaw integration

- Research pages: Research Intelligence, Topic Monitor, Ticker Research, Intelligence Hub, Overnight Brief
- AI Analyst and CIO Dashboard
- Tax & Lots, Rebalance, and Retirement pages
- Proposal lifecycle inspector
- Execution-readiness endpoint
- Backtesting with source-aware run types
- LLM Review Coverage label clarity
- Complete backtest classification coverage
- ATM audit showing automated trading is working correctly but blocked when appropriate
- Human-in-the-loop execution and fail-closed safety principles

Hermes should use those existing capabilities, not duplicate them.

4. Operating Model

4.1 Trade AI Owns

- Proposal creation
- Proposal enrichment
- Proposal lifecycle state
- ATM approval logic
- Execution-readiness checks
- Broker and paper-trade adapters
- Journal records
- Backtesting database
- Trade LLM review records
- Stop management
- Risk gates
- Automated-trading safety
- Command Center UI
- System health monitoring
- Model routing policy
- Google Drive sync pipeline

4.2 Hermes Owns

- Research synthesis
- Ticker dossiers
- News and transcript reframing
- Incubator research
- Trade reflection
- Proposal challenge memos
- Missed-opportunity analysis
- Portfolio rotation research
- Retirement research
- Tax and lot research packets

- Strategy hypothesis generation
- Long-term memory and lessons
- Recommendation queue
- Daily/weekly/monthly advisory briefs

4.3 Claude Code Owns

- Code implementation
- Documentation updates
- Schema or config changes when approved
- Controlled validation
- Commits and Drive sync
- Rollback documentation

5. Hermes Safety Boundary

Hermes must never:

- Place orders
- Call broker submit endpoints
- Approve proposals
- Reject proposals
- Expire proposals
- Modify paper_trades
- Modify real trade records
- Modify journal rows
- Modify proposal lifecycle status
- Change `.env`
- Change cron
- Change model routing
- Change production strategy config directly
- Run arbitrary shell commands against production
- Create a second trading worker

Hermes may:

- Read from approved APIs, exports, docs, and snapshots
- Write advisory JSONL memory
- Write Markdown reports
- Write recommendation queue items
- Draft Claude Code implementation plans
- Flag missing evidence
- Ask for operator review
- Summarize and research

6. The Six Hermes Pods

The full v4 design groups Hermes into six pods. Pods make the system easier to operate than a flat list of agents.

Pod	Purpose	Main Outputs
Research Intelligence Pod	Research tickers, news, transcripts, articles, and external narratives	Ticker dossiers, news scores, transcript briefs
Trade Lifecycle Pod	Review trades, proposals, open positions, missed opportunities, and thesis decay	Reflections, challenge memos, missed-opportunity reports
Portfolio Planning Pod	Research holdings, rotation, retirement, taxes, dividends, ETFs, and rebalancing	Rotation memos, tax-lot watchlists, retirement packets
Strategy and Experiment Pod	Turn lessons into scientific one-variable strategy experiments	Hypotheses, experiment backlog, regime reports
Operations and Governance Pod	Research internal issues, tickets, dashboards, freshness, and documentation gaps	Issue digests, freshness warnings, UI truth audits
Coordinator and Memory Pod	Orchestrate agents, schedule work, maintain memory, score recommendations	Daily brief, weekly review, memory index, outcome scoring

7. Hermes Agent Roster v4

The target design is 24 logical Hermes agents. They can start as workflows in one sidecar and later split into scheduled services.

7.1 Agent 1 - Chief Hermes Coordinator

Purpose: Coordinate all Hermes agents and produce the operator-facing picture.

Jobs:

- Decide which agents run and when
- Prevent duplicate work
- Maintain daily/weekly/monthly schedules
- Route tasks to the right model tier
- Summarize agent findings
- Escalate urgent items
- Maintain the Hermes operating state

Outputs: Daily Hermes Brief, weekly review, monthly review, coordinator log, priority queue.

7.2 Agent 2 - Ticker Research Agent

Purpose: Build living ticker dossiers.

Researches: Fundamentals, technicals, analyst ratings, price targets, sector context, news, prior Trade AI history, prior rejected proposals, backtest evidence, current setup readiness.

Outputs: Ticker dossier, bull/bear case, catalyst score, risk score, missing evidence, next action.

7.3 Agent 3 - News Research Agent

Purpose: Classify and prioritize news.

Researches: News articles, related news, article age, source quality, repeated stories, ticker linkage, catalyst relevance.

Outputs: Actionable news, noise list, thesis-change alerts, ticker-linked summaries.

7.4 Agent 4 - YouTube and Transcript Research Agent

Purpose: Convert long-form video and transcripts into trade intelligence.

Researches: YouTube search results, YouTube transcripts, earnings transcripts, interviews, investor days, conference transcripts.

Outputs: Transcript brief, ticker implications, management tone, catalyst/risk extraction, follow-up questions.

7.5 Agent 5 - Research Reframer Agent

Purpose: Translate raw research into trade implications.

Researches: Articles, transcripts, analyst notes, tickets, internal notes.

Outputs: Why-this-matters summary, bull case, bear case, timing relevance, confidence score.

7.6 Agent 6 - Source Credibility Agent

Purpose: Score whether research can be trusted.

Researches: Source type, original reporting, recency, repeated content, source history, relevance to ticker.

Outputs: Source credibility score, stale-source warnings, duplicate-story warnings, confidence adjustment.

7.7 Agent 7 - Incubator Research Agent

Purpose: Review pre-trade ideas in the incubator and percolator.

Researches: Incubator names, percolator candidates, watchlists, topics, news, transcript findings, catalyst status, sector context.

Outputs: Promote / hold / drop / needs evidence, research completeness score, missing-data checklist.

7.8 Agent 8 - Proposal Challenge Agent

Purpose: Independently challenge Trade AI proposals before action.

Researches: Proposal enrichment, lifecycle inspector, ticker dossier, news, backtests, similar trades, risk state.

Outputs: support / wait / challenge / needs evidence / reject recommendation / operator review required.

7.9 Agent 9 - All-Trade Reflection Agent

Purpose: Review all trades across paper and real modes.

Researches: Closed trades, open-to-close lifecycle, journal, original proposal, news around entry/exit, stop behavior, MFE/MAE.

Outputs: setup score, exit score, stop score, mistake category, learning note, one-variable experiment candidate.

7.10 Agent 10 - Open Trade Watch Agent

Purpose: Monitor open positions for thesis deterioration or opportunity.

Researches: Holdings, open trades, stop alerts, latest news, analyst changes, sector movement, portfolio heat.

Outputs: hold / review / reduce / exit-watch recommendation, evidence summary, alert priority.

7.11 Agent 11 - Missed Opportunity Agent

Purpose: Learn from what Trade AI did not take.

Researches: Rejected proposals, expired proposals, missed proposals, incubator names not promoted, later price movement, later news.

Outputs: false-negative score, block reason, whether the block was correct, suggested gate test.

7.12 Agent 12 - Thesis Decay Agent

Purpose: Detect when an original thesis is weakening.

Researches: Original catalyst, latest news, analyst downgrades, sector weakness, price action, expired catalysts, negative articles.

Outputs: thesis intact / weakened / broken, evidence, recommended review urgency.

7.13 Agent 13 - Strategy Hypothesis Agent

Purpose: Generate scientific one-variable strategy experiments.

Researches: Trade reflections, missed opportunities, backtests, journal, strategy performance, prior experiments.

Outputs: current variable, proposed variable, evidence, expected impact, test plan, rollback criteria.

7.14 Agent 14 - Backtest and Journal Consistency Agent

Purpose: Check whether learning data is clean.

Researches: Backtest rows, trade LLM reviews, journal records, trade_transactions, proposal links, run_type labels.

Outputs: data-quality report, source mismatch report, review coverage warnings.

7.15 Agent 15 - Market Regime and Macro Research Agent

Purpose: Contextualize trades and ideas by regime.

Researches: indexes, VIX, sector ETFs, breadth, rates, macro calendar, risk regime, portfolio heat.

Outputs: regime label, strategy compatibility, macro warning, sector pressure report.

7.16 Agent 16 - Portfolio Rotation Research Agent

Purpose: Research holdings and possible rotation opportunities.

Researches: holdings, attribution, returns, dividends, analyst changes, sector rotation, correlation, forecast.

Outputs: rotation watchlist, hold/review/reduce notes, replacement candidates.

7.17 Agent 17 - Dividend and Income Research Agent

Purpose: Research dividend sustainability and income quality.

Researches: dividends, payout trends, ETF/fund income, AGNC-like income positions, dividend cuts, income gaps.

Outputs: dividend concern notes, income-quality score, replacement research list.

7.18 Agent 18 - Retirement Research Agent

Purpose: Research retirement-account holdings and long-term suitability.

Researches: IRA/401k holdings, ETFs, funds, long-term allocation, sector exposure, risk, income, retirement roadmap.

Outputs: retirement review, ETF/fund replacement candidates, risk concentration notes, long-term suitability questions.

7.19 Agent 19 - Tax and Lots Research Agent

Purpose: Prepare tax-lot and tax-planning research for operator/CPA review.

Researches: tax lots, realized/unrealized gains, holding periods, wash-sale indicators if available, dividends, account type.

Outputs: tax-loss watchlist, holding-period alerts, CPA question list, rebalance/tax impact notes.

Important: Hermes produces research only. It does not provide final tax advice.

7.20 Agent 20 - Rebalance Research Agent

Purpose: Research rebalancing opportunities and risks.

Researches: current allocation, target allocation, concentration, sector drift, tax impact, retirement suitability, portfolio heat.

Outputs: rebalance candidates, risk notes, income impact, tax-aware considerations.

7.21 Agent 21 - Internal Ticket and Issue Research Agent

Purpose: Research internal operational issues and tickets.

Researches: tickets, Telegram alerts, Claude Code reports, health-agent findings, failed jobs, unresolved TODOs.

Outputs: issue digest, recurring failure report, priority list, next-session task list.

7.22 Agent 22 - Data Freshness Critic

Purpose: Prevent stale data from driving bad conclusions.

Researches: freshness registry, pipeline timestamps, last successful runs, stale dashboards, missing enrichment.

Outputs: stale-data warnings, confidence downgrades, remediation suggestions.

7.23 Agent 23 - Dashboard Truth Auditor

Purpose: Review UI screenshots and labels for operator misunderstanding.

Researches: screenshots, Playwright crawls, page payloads, labels, warnings, counts.

Outputs: misleading-label warnings, UI comprehension issues, recommended clarity fixes.

7.24 Agent 24 - System Memory and Lessons Agent

Purpose: Maintain durable memory and outcome scoring.

Researches: all Hermes outputs, operator decisions, outcomes, Claude Code reports, session summaries, lessons learned.

Outputs: memory index, recommendation outcomes, do-not-repeat list, next-session memory notes.

8. Research Objects Hermes Should Maintain

8.1 Living Ticker Dossier

Each important ticker should have a persistent dossier:

```
ticker: XYZ

last_updated: timestamp

current_status: incubator | proposal | open_trade | closed_trade | avoid | watch

thesis: text

bull_case: text

bear_case: text

latest_news: []

latest_transcripts: []

analyst_context: {}

sector_context: {}

trade_ai_history: []

proposal_history: []
```

```
backtest_context: []

open_questions: []

next_action: text

confidence: 0.0
```

8.2 Research Debt Record

Hermes should track what is missing:

- no recent news
- no transcript
- no analyst data
- no sector comparison
- no backtest sample
- no historical Trade AI history
- contradictory sources unresolved
- stale data
- weak catalyst

8.3 Reason-We-Did-Not-Trade Record

For promising ideas not traded:

- no catalyst
- stale quote
- bad market regime
- poor risk/reward
- missing enrichment
- operator skipped
- too late
- earnings risk
- portfolio heat

Hermes should later check whether that reason was correct.

8.4 Operator Decision Memory

Hermes should remember John's decisions:

- approved
- rejected
- deferred
- requested more evidence
- overrode Hermes
- agreed with Hermes
- ignored recommendation

Later, Hermes checks outcomes and adapts.

8.5 Recommendation Queue

File-first implementation:

```
data/hermes_memory/recommendation_queue.jsonl
```

Future DB table:

```
hermes_recommendation_queue
```

Fields:

- recommendation_id
- agent_name
- symbol
- strategy
- source_type
- source_ids
- recommendation_type
- summary
- evidence
- confidence
- risk_level
- required_operator_action
- suggested_next_step
- expiration_time
- status
- outcome

9. Near-24/7 Operating Schedule

9.1 Lightweight Always-On Loop

Cadence: every 15-30 minutes.

Jobs:

- Check new news/articles/transcripts
- Check incubator changes
- Check proposal changes
- Check open trade alerts
- Check system alerts
- Update memory index
- Queue deeper research

Model: gemma3:4b for light summaries, gemma3:12b for moderate analysis.

9.2 Market Hours Loop

Jobs:

- Open trade watch
- Proposal challenge review
- News catalyst updates
- Ticker risk check
- Intraday issue detection
- Thesis decay detection

Model: gemma3:12b, fallback gemma3:4b.

9.3 After-Close Loop

Jobs:

- All-trade reflection
- Proposal outcome review
- Missed opportunity review
- Journal/backtest comparison
- Daily Hermes Brief

Model: gemma3:12b.

9.4 Overnight Deep Loop

Jobs:

- YouTube transcript analysis
- Long article synthesis
- Incubator deep review
- Retirement/portfolio research
- Tax-lot watchlist generation
- Strategy hypothesis generation

Model: Gemma4 31B via llama.cpp for selected batches; fallback to gemma3:12b.

9.5 Weekly Loop

Jobs:

- Portfolio rotation review
- Retirement review
- Tax/rebalance watchlist
- Strategy lessons
- Missed opportunity review
- Top research findings

9.6 Monthly Loop

Jobs:

- CIO-style review
- Strategy promotion/demotion
- Research pipeline review
- Tax and lot planning watchlist
- Retirement suitability review
- Model performance review
- Hermes recommendation outcome review

10. Model Routing Strategy

Hermes should be model-router aware. It should request an analysis tier; the Trade AI model policy should decide the actual engine.

Workload	Preferred Model
short summaries	gemma3:4b
normal ticker research	gemma3:12b
proposal challenge	gemma3:12b
trade reflection	gemma3:12b
incubator research	gemma3:12b
transcript synthesis	gemma3:12b or Gemma4 31B off-hours
weekly/monthly deep review	Gemma4 31B via llama.cpp
external narrative challenge	optional xAI/Grok later

Disabled / not production:

- qwen3:14b
- Gemma4 e2b/e4b
- gemma3:27b GPU

10.1 Grok / xAI Position

Grok/xAI should not be the default Hermes brain.

It can be added later as an External Challenger Agent for:

- broad market narrative
- social/X-aware sentiment
- outside-view macro review
- high-value ticker challenge
- weekly/monthly alternative perspective

Controls:

- secrets in vault or `.env`
- no hardcoded keys
- cost limits
- rate limits
- sanitized prompts
- no broker credentials

- no account numbers
- advisory output only

11. Command Center Integration

Hermes should eventually be visible in:

- AI Analyst
- CIO Dashboard
- Research Intelligence
- Topic Monitor
- Ticker Research
- Intelligence Hub
- Incubator
- Strategy Desk
- Proposal Lifecycle Inspector
- Journal Reports
- Backtesting
- Tax & Lots
- Rebalance
- Retirement
- System Health
- Agent Pipeline

Suggested UI cards:

- Hermes Daily Brief
- Hermes Research Queue
- Hermes Ticker Dossiers
- Hermes Incubator Watch
- Hermes News Reframes
- Hermes Transcript Findings
- Hermes Trade Lessons
- Hermes Missed Opportunities
- Hermes Strategy Hypotheses
- Hermes Portfolio Rotation
- Hermes Retirement Watch
- Hermes Tax/Lot Watch
- Hermes Internal Issues
- Hermes Memory Outcomes

12. First Build Target

The first build should be read-only and research-first.

12.1 Phase 1 Agents

Start with five:

1. Chief Hermes Coordinator
2. Ticker Research Agent
3. News Research Agent
4. Incubator Research Agent
5. All-Trade Reflection Agent

12.2 Phase 1 Inputs

- latest 25 closed trades across all modes
- current open trades
- current incubator items
- latest 50 news articles
- latest related news
- latest YouTube transcripts if available
- latest rejected/expired proposals
- latest missed proposals
- latest portfolio holdings
- latest retirement holdings
- latest tax/rebalance watchlist
- latest internal tickets

12.3 Phase 1 Outputs

- Daily Hermes Research Brief
- Top ticker risks/opportunities
- Top incubator promotion/drop candidates
- Top trade lessons
- Top missed opportunities
- One one-variable strategy hypothesis
- Research debt list
- Memory files written
- No DB writes
- No proposal mutations
- No execution actions

13. Execution Plan Preview

Step 1 - Hermes Compatibility Audit

Before installation:

- Verify whether Hermes is already installed
- Determine how Hermes stores memory
- Determine whether Hermes can use local models
- Determine whether Hermes calls external APIs by default
- Determine whether Hermes can run project-scoped
- Determine whether Hermes can be sandboxed
- Determine whether Hermes supports multiple agents/workflows

- Determine whether it conflicts with Claude Code
- Determine install/rollback behavior

No install without approval.

Step 2 - Sidecar Directory Skeleton

Create:

```
docs/hermes/

data/hermes_memory/

logs/hermes/

config/hermes/
```

Step 3 - Read-Only Connectors

Build exports or read-only API pulls for:

- trades
- proposals
- incubator
- research
- news
- transcripts
- backtests
- journal
- risk
- portfolio
- tax lots
- retirement
- system tickets

Step 4 - First Manual Hermes Run

Run the five Phase 1 agents manually.

Step 5 - Operator Review

Review:

- source accuracy
- hallucinations
- usefulness
- missing context
- actionability
- memory quality

Step 6 - Add Schedule

Only after useful output:

- 15-30 minute lightweight polling
- after-close daily report
- overnight deep research
- weekly/monthly reports

Step 7 - Add Command Center Cards

Expose outputs in UI.

Step 8 - Add External Challenger

Add xAI/Grok only after local Hermes memory loop is proven.

14. Success Metrics

Hermes is successful if it:

- Finds missed risks before they damage trades
- Finds missed opportunities Trade AI skipped
- Produces useful ticker dossiers
- Improves incubator research quality
- Reduces repeated mistakes
- Produces measurable strategy hypotheses
- Tracks whether its own recommendations worked
- Avoids hallucinated source IDs
- Does not mutate trading state
- Does not create duplicate execution paths
- Improves John's decision speed and confidence

15. Final v4 Recommendation

Build Hermes as a near-24/7 research, memory, and challenge sidecar for Trade AI.

Use the six-pod structure and 24 logical agents as the target design.

Start with five agents and file-based memory.

Use local LLMs first.

Use Gemma4 31B only for off-hours deep research.

Add Grok/xAI later only as an external challenger.

Keep Trade AI as the only execution authority.

Make Hermes self-learning by recording every recommendation, John's decision, and the later outcome.

The first real implementation task should be a Hermes compatibility audit and sidecar install plan, not installation.